

HW 2 - Warren montes

$$3.3) P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$E[h(X)] = \sum_{\text{all } x} h(x) p(x)$$

$$1. P(X=x) = \binom{3}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{3-x}$$

$\Downarrow$

$$P(X=x) = \binom{3}{x} \left(\frac{1}{2}\right)^3 = \frac{\binom{3}{x}}{8}$$

Possible $X = 0, 1, 2, 3$

$$P(X=0) = \binom{3}{0} / 8 = 1/8$$

$$P(X=1) = \binom{3}{1} / 8 = 3/8$$

$$P(X=2) = \binom{3}{2} / 8 = 3/8$$

$$P(X=3) = \binom{3}{3} / 8 = 1/8$$

$$\text{Using } E[X] = \sum_x x P(X=x)$$

$$E[X] = 0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 2\left(\frac{3}{8}\right) + 3\left(\frac{1}{8}\right)$$

$$= 0 + 3/8 + 6/8 + 3/8$$

$$= 3/2$$

$$= 1.5$$